

Telco Customer Churn Prediction

Case Study Documentation — AA DSS110

Dataset: IBM Telco Customer Churn (Kaggle: [blastchar/telco-customer-churn](#)) · Problem type: supervised binary classification · Companion notebook: [Telco_Churn_Case_Study.ipynb](#)

Overview

This document answers the six case-study steps for the proposal *Telco Customer Churn Prediction*: collect data, preprocess and clean (with exploratory data analysis), know the target variable, train a model, evaluate it, and predict on unseen data. As instructed, most of the effort goes into EDA; the modelling code follows the standard scikit-learn workflow adapted to this data. All numbers below come from the executed notebook.

Summary of results

- **Data:** 7,043 customers × 21 columns; 26.5% churned (imbalanced).
- **Data quality:** one issue — 11 blank `TotalCharges` values (all customers with tenure 0). No duplicates, no other nulls, no outliers.
- **Top churn drivers:** month-to-month contract (42.7% churn vs 2.8% on two-year), fiber-optic internet (41.9%), electronic-check payment (45.3%), tenure under 12 months (47.4%), and no security/tech-support add-ons.
- **Best model by test ROC-AUC:** Random Forest (0.846); all models land within 0.005 of each other (0.842–0.846).

Step 1 — Collect data

Choice of dataset. The IBM Telco Customer Churn dataset is a classification dataset: each row describes one telecom customer and the label `Churn` (Yes/No) records whether they left in the last month. This matches the proposal's binary-classification problem.

Source. Kaggle ([blastchar/telco-customer-churn](#)), which republishes IBM sample data. Kaggle requires a login, so the identical file was downloaded from IBM's public GitHub repository ([IBM/telco-customer-churn-on-icp4d](#)) and saved as `Telco-Customer-Churn.csv`.

Structure. 7,043 rows × 21 columns: 3 numeric (`tenure`, `MonthlyCharges`, `TotalCharges`), 1 numeric flag (`SeniorCitizen`), 16 categorical features including demographics (`gender`, `partner`, `dependents`), services (`phone`, `internet`, `add-ons`, `streaming`), account data (`contract`, `paperless billing`, `payment method`), one ID column and the target.

Step 2 — Preprocess and clean data (EDA and wrangling)

2.1 Data-quality audit

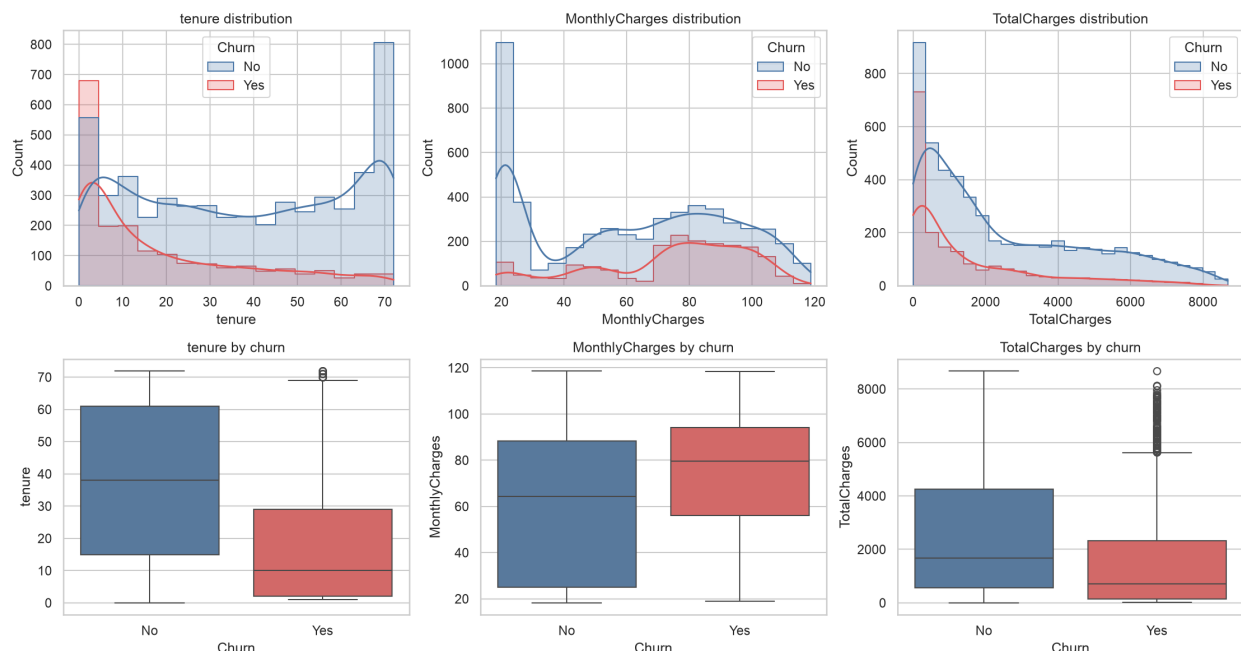
Check	Result
Duplicate rows / duplicate customer IDs	0 / 0
Null values	0
Non-numeric TotalCharges	11 blank strings, all with tenure = 0 (new customers, not yet billed)
IQR outliers (tenure, MonthlyCharges, TotalCharges)	0 in each; values are bounded and plausible
Redundant labels	"No internet service" / "No phone service" repeat information held in InternetService / PhoneService

2.2 Data wrangling steps

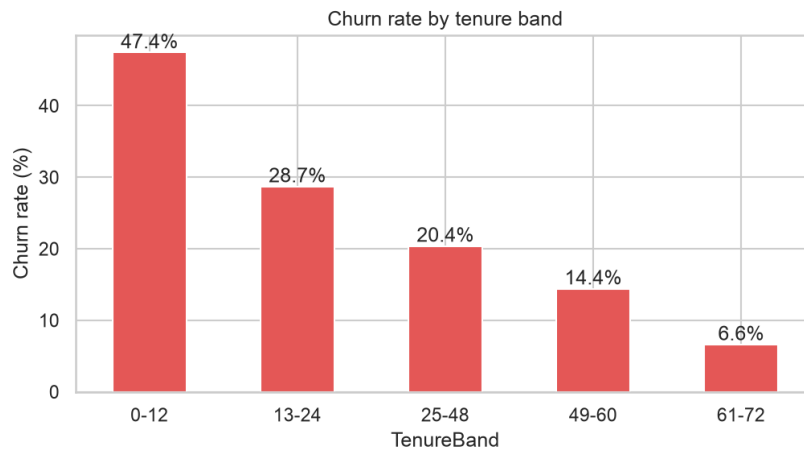
- TotalCharges converted from text to numeric; the 11 blanks were set to 0 because tenure is 0 (no charges accrued) — dropping them would also be valid (0.16% of rows).
- Dropped customerID (identifier with no predictive value).
- Recoded SeniorCitizen from 0/1 to No/Yes for readable EDA; merged "No internet/phone service" into "No" for add-on columns.
- Categorical variables one-hot encoded (first level dropped); numeric variables standardised (StandardScaler) inside the modelling pipeline so statistics come from training data only.
- No outliers were found, so no rows were removed.

2.3 Exploratory data analysis

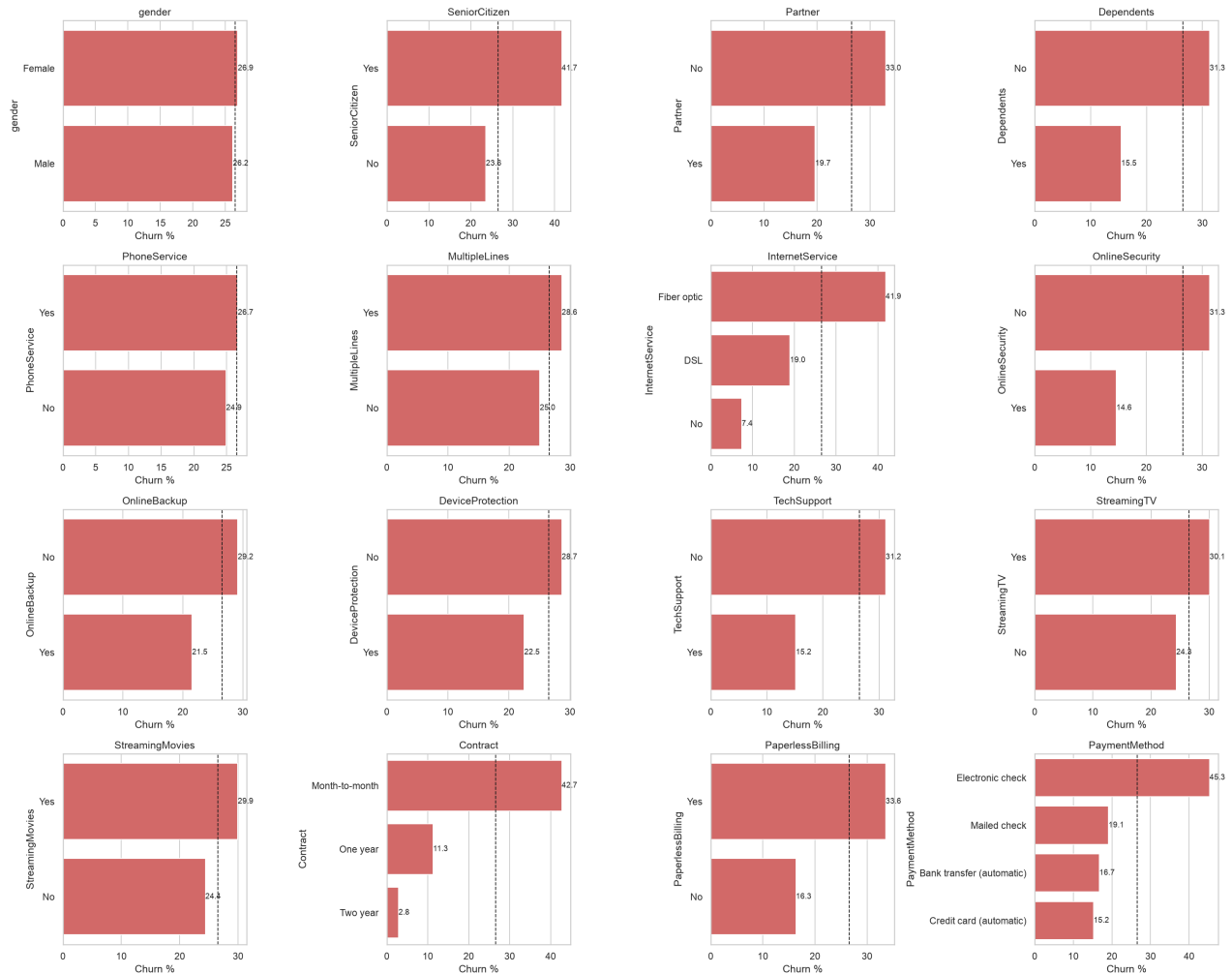
Numeric features. Churners have a far shorter tenure (median 10 vs 38 months) and higher monthly charges (median \$79.65 vs \$64.43). Their total charges are lower (median \$703.55 vs \$1,679.53) simply because they leave early. Mann-Whitney U tests find all three differences significant ($p < 1e-50$).



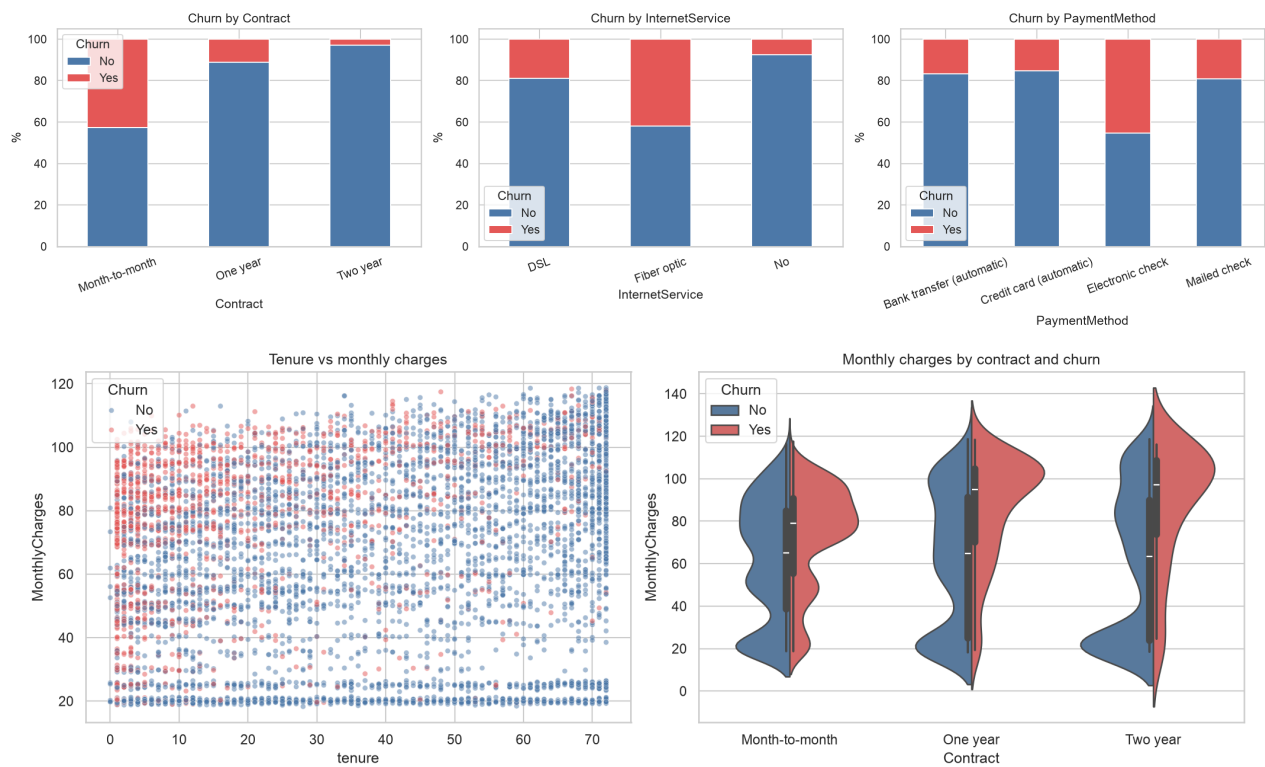
Tenure. Churn falls steadily with loyalty: 47.4% in the first 12 months, 28.7% in months 13–24, 20.4% in months 25–48, 14.4% in months 49–60 and 6.6% beyond 60 months.



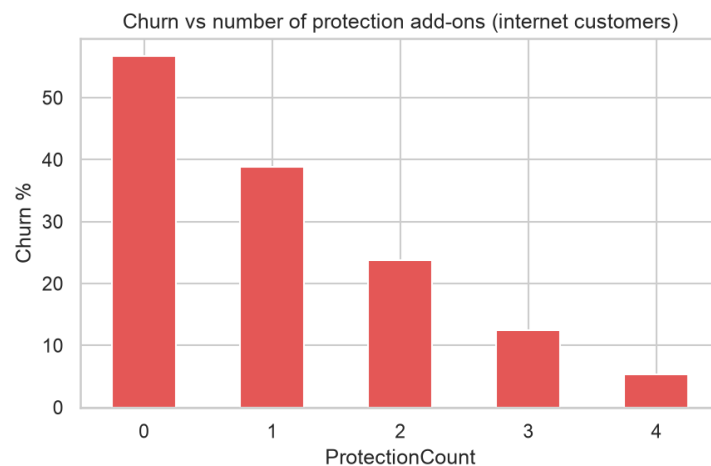
Categorical features. Dashed lines mark the overall churn rate (26.5%). Bars to the right of the line indicate above-average risk.



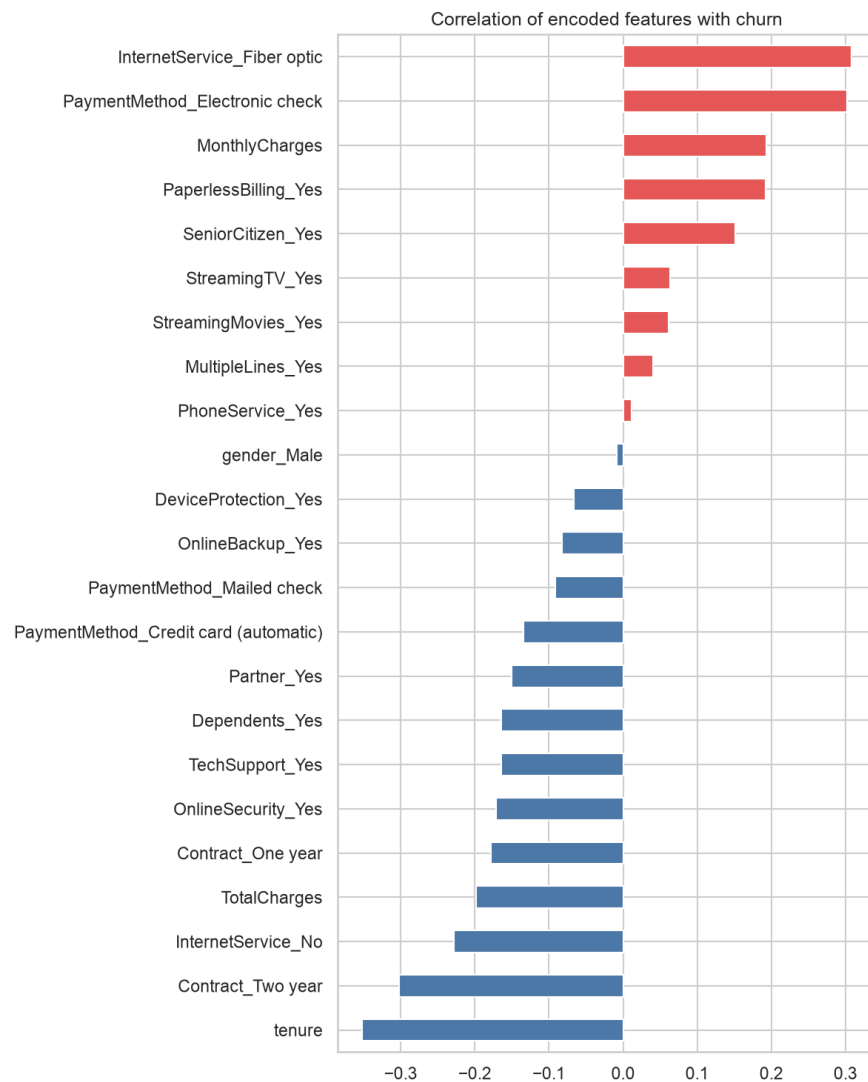
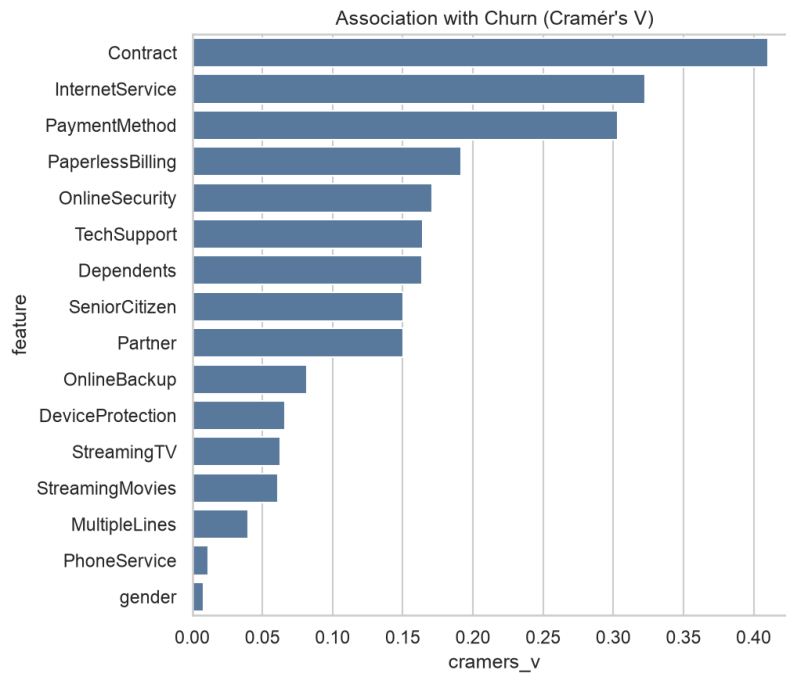
Key drivers. Month-to-month customers churn at 42.7% against 11.3% (one-year) and 2.8% (two-year). Fiber-optic customers churn at 41.9% versus 19.0% for DSL and 7.4% for no internet. Electronic-check payers churn at 45.3%, about three times the rate of automatic payment methods (15.2–16.7%).

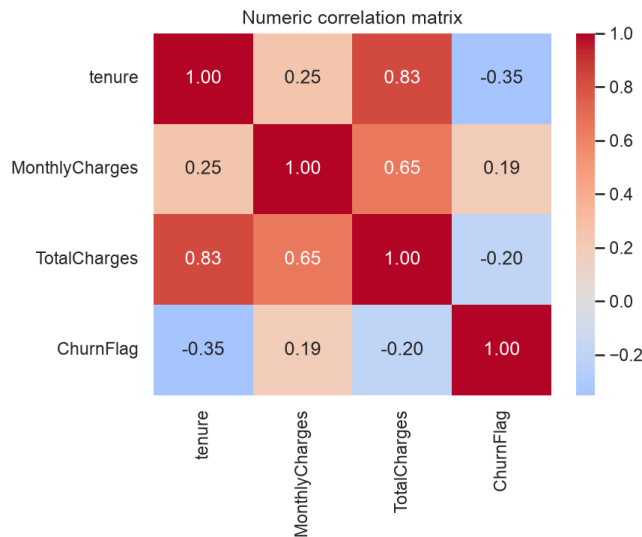


Protective services. Among internet customers, churn drops from 56.7% with no protection add-on (security, backup, device protection, tech support) to 5.3% with all four.



Statistical association. Chi-square tests with Cramér's V (effect size) rank *Contract* ($V = 0.41$), *InternetService* (0.32) and *PaymentMethod* (0.30) highest, followed by *PaperlessBilling*, *OnlineSecurity* and *TechSupport*. *gender* ($p = 0.49$) and *PhoneService* ($p = 0.34$) are not significantly related to churn.





Correlation. Tenure and TotalCharges are strongly correlated (multicollinearity), which regularised or tree-based models handle well. Tenure has the strongest negative correlation with churn, and fiber optic / electronic check / month-to-month the strongest positive ones.

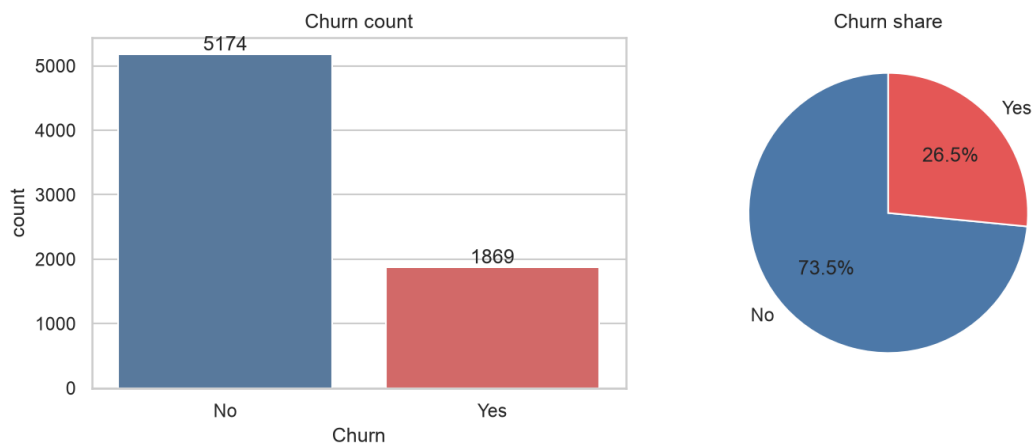
EDA conclusions

- Churn concentrates in new customers on month-to-month contracts with fiber-optic internet, paying by electronic check and without security or tech-support add-ons.
- Long-tenure customers, two-year contracts and customers with several protection add-ons rarely leave.
- Gender and phone service carry almost no signal.

Step 3 — Know your target variable

The target is Churn (Yes = left, No = stayed). The classes are imbalanced: **5,174 No (73.46%)** and **1,869 Yes (26.54%)**, a ratio of about 2.8 : 1.

Implication. A model that always predicts "No" would be about 73.5% accurate and catch zero churners, so accuracy alone is misleading. The project therefore uses a stratified train/test split (both sets keep 26.5% churn), class weighting during training, and precision, recall, F1 and ROC-AUC for evaluation.



Step 4 — Train a model

Split. 80/20 stratified split (random_state = 42): 5,634 training rows and 1,409 test rows, each with a 26.5% churn rate. The test set is held out and used only for evaluation.

Pipeline. StandardScaler on the 3 numeric features + one-hot encoding on the 16 categorical features → classifier. Imbalance is handled with class weights.

Model	Parameters
Logistic Regression	C = 1.0, penalty = L2, class_weight = 'balanced', max_iter = 1000
Random Forest	n_estimators = 300, max_depth = 8, min_samples_leaf = 5, class_weight = 'balanced'
Gradient Boosting	n_estimators = 200, learning_rate = 0.05, max_depth = 3, balanced sample weights

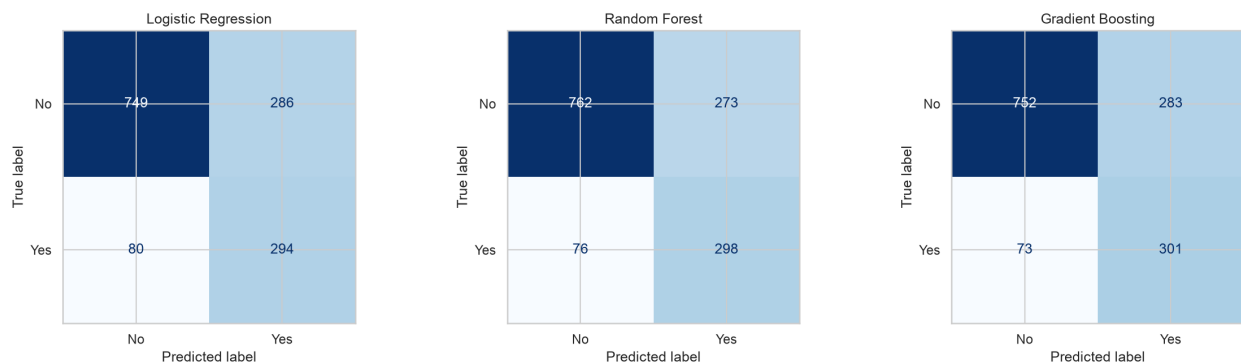
Logistic Regression was additionally tuned with 5-fold stratified cross-validation over $C \in \{0.01, 0.1, 1, 10\}$ using ROC-AUC; the best value was $C = 10$ (CV ROC-AUC 0.846).

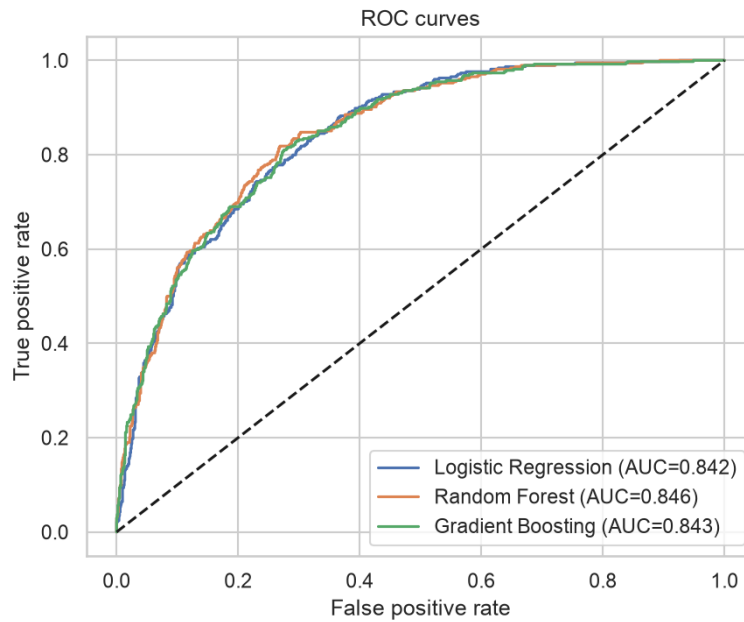
Step 5 — Evaluate the model

Classification metrics on the 1,409-customer test set (threshold 0.5; "Churn = Yes" is the positive class):

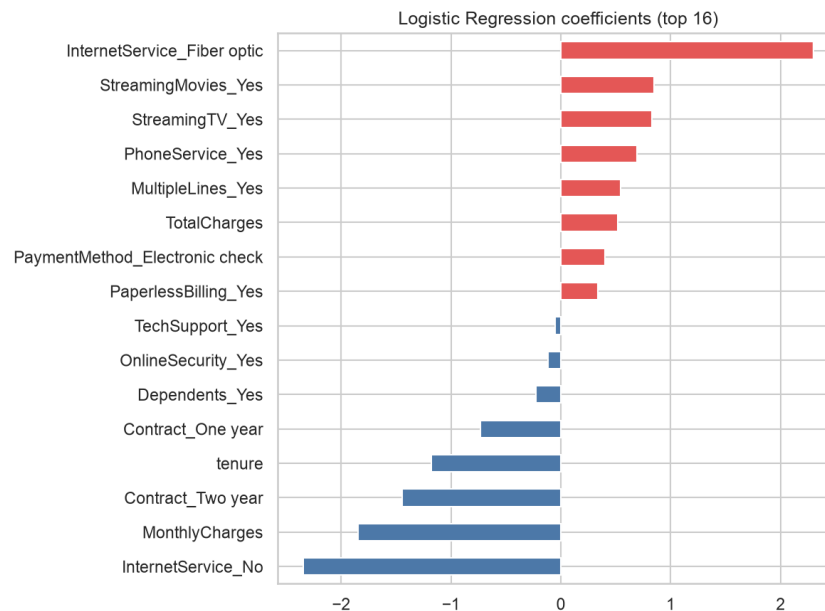
Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.740	0.507	0.786	0.616	0.842
Random Forest	0.752	0.522	0.797	0.631	0.846
Gradient Boosting	0.747	0.515	0.805	0.628	0.843
Logistic Regression (tuned)	0.740	0.507	0.783	0.616	0.841

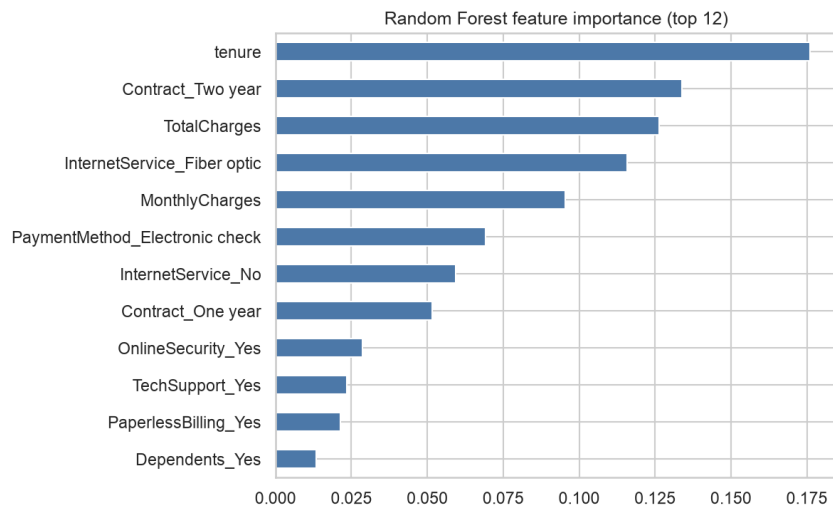
Interpretation. All models perform similarly: ROC-AUC 0.841–0.846, i.e. they rank a random churner above a random non-churner about 84% of the time. Random Forest is marginally best on accuracy, F1 and AUC; Gradient Boosting has the highest recall (80.5%). Because of class weighting, accuracy (about 74–75%) is deliberately traded for recall (about 79–80%): roughly four of five churners are caught, but precision is about 51–52%, so about half of the customers flagged as at-risk will not actually leave. That is an acceptable trade-off if a retention offer is cheap relative to the value of a lost customer. Tuning Logistic Regression did not improve the test score.





Feature influence. In Logistic Regression, tenure, two-year/one-year contracts and no-internet service lower churn probability, while fiber-optic internet, month-to-month contracts, electronic-check payment and higher monthly charges raise it. The Random Forest ranks tenure, TotalCharges, MonthlyCharges, contract type and fiber-optic internet as most important, consistent with the EDA.





Step 6 — Use the model to predict on new data

The 20% test split was never used during training or tuning, so predictions on it show how the model behaves on unseen customers. Metrics in Step 5 are exactly this. The notebook also scores sample customers with the tuned Logistic Regression. Two hand-built new customers:

Profile	Predicted churn probability
New customer: tenure 2 months, month-to-month, fiber optic, electronic check, \$95/month, no add-ons	87.4% (high risk)
Loyal customer: tenure 60 months, two-year contract, DSL, automatic bank transfer, all protection add-ons	2.3% (low risk)

Among the first eight test customers, the model flagged four as likely churners; none of those four had actually left (an illustration of the precision limitation above), and it correctly scored the long-tenure two-year customers as low risk (6–13%).

Conclusion and recommendations

- Customer churn can be predicted reasonably well from account data (ROC-AUC about 0.84, recall about 80% on churners), but precision is limited (about 51%).
- Random Forest is best on raw metrics by a very small margin; Logistic Regression is nearly as good and far easier to explain to retention teams.
- Business actions suggested by the EDA: offer contract-upgrade incentives to month-to-month customers in their first year; promote automatic payment instead of electronic check; bundle online security and tech support, especially with fiber-optic plans.
- Limitations: a single snapshot dataset (no time dimension), a default 0.5 threshold, and only light tuning. Future work: threshold selection by retention-cost analysis, SMOTE/resampling comparison, SHAP explanations, and the Streamlit app proposed in the project proposal.

Deliverables: `Telco_Churn_Case_Study.ipynb` (executed notebook), this PDF, `Telco-Customer-Churn.csv`, `figures/`.